

Asmahan Matooq S Al Safwan, PhD

Education Researcher | Physics & Science Education | STEM Education | Early Childhood Pedagogy

[Academic website](#) • [Google Scholar](#) • [ORCID](#) • [Email](#)

Academic profile

Education researcher with a disciplinary foundation in physics and extensive experience in secondary physics teaching. Academic research began in applied physics, nanotechnology and materials science before moving within Curtin into science education. Doctoral research examined Saudi secondary students' and teachers' understandings of thermal concepts, with particular attention to conceptual understanding, conceptual change, language, culture and curriculum. Current research extends these interests in how learners construct and transform understanding into early childhood education, narrative pedagogy and the development of Guardian Pedagogy.

Research interests

- Science and physics education
- Conceptual understanding and conceptual change
- Early childhood pedagogy and curriculum
- STEM education and teacher learning
- Curriculum, language and culture in learning
- Narrative inquiry and story-based learning

Education

2019 Doctor of Philosophy — Curtin University, Australia

Degree conferred 6 December 2019. Formal Curtin course registration: Doctor of Philosophy – Applied Physics; enrolling school: School of Education. Thesis: Saudi Female High School Students' and Teachers' Understandings of Thermal Concepts. Doctoral research situated in physics/science education and conceptual change; internationally examined, with both examiners recommending passage with relatively minor revisions.

2013 Master of Science (Physics) — Curtin University of Technology, Australia

Degree conferred 11 January 2013.

1990 Bachelor of Science (Physics) — King Faisal University, College of Science in Dammam, Saudi Arabia

Degree awarded 3 October 1990.

Research trajectory and current programme

Physics, science education and conceptual change

Research developed from applied physics, nanotechnology and materials science into physics and science education, focusing on how learners understand thermal concepts and how language, culture, curriculum and scientific terminology shape conceptual understanding and change.

Guardian Pedagogy (2026–present)

A current conceptual research programme extending a longstanding interest in learning and meaning-making into early childhood contexts. The programme examines how narrative, inquiry, relationships, culture and curriculum can support enduring and renewable learning.

- Story Affordances and narrative renewability

- The Guardian Inquiry Cycle as a recursive process model
- Human Questions as continuity-bearing structures in inquiry
- Cultural, relational and ethical dimensions of story-based curriculum

Peer-reviewed and scholarly publication

Albetran, H., Alsafwan, A., Haroosh, H., Dong, Y., & Low, I. M. (2013). X-Ray Diffraction Study on the In-Situ Crystallization Kinetics in Electrospun PVP/TiO₂ Nanofibers. *Nanostructured Materials and Nanotechnology VII*, 35–49. Wiley. DOI: 10.1002/9781118807828.ch4

Doctoral thesis

Al Safwan, A. (2018). Saudi Female High School Students' and Teachers' Understandings of Thermal Concepts. Curtin University.

The study investigated conceptual understanding of thermal phenomena in a Saudi educational context and examined the roles of language, culture and curriculum in science learning. Both international examiners recommended passage with relatively minor revisions.

Selected conference contributions

2019 An International Comparison of Students' Understanding of Thermal Concepts prior to Instruction — American Educational Research Association (AERA) Annual Meeting, SIG–Science Teaching and Learning

Peer-reviewed conference proposal; co-authors David F. Treagust and Marjan Zadnik.

2018 An Arabic Culture Model of Conceptual Change — Second Inter-Regional Research Conference on Science and Mathematics Education, American University of Beirut, Lebanon

Conference paper presented and discussed, 8–9 December 2018; co-authors David F. Treagust and Marjan Zadnik.

2018 The Arab Culture Model of Conceptual Change — 2nd International Conference on STEM Education (ICSTEM 2018), Kuala Lumpur, Malaysia

Poster presentation, 2–3 October 2018; co-authors David F. Treagust and Marjan Zadnik.

2017 Do Students in Developing Countries Understand Science Differently? — 5th IPCSM / ICSTEM2017, Universiti Pendidikan Sultan Idris, Malaysia

Oral presentation, 7–9 October 2017.

2015 Female High School Students Understanding of Thermal Concepts — Australian Association for Research in Education (AARE)

Abstract-refereed conference contribution.

Teaching and educational experience

Early childhood educator / manager — Olive Grove Centre of Early Learning, Western Australia | Current

- Early childhood curriculum planning, documentation, reflective practice and educator support.
- Practice aligned with the Early Years Learning Framework, National Quality Standard and Australian early childhood regulatory context.
- Focus on culturally responsive practice, family partnerships, inclusion and Aboriginal and Torres Strait Islander perspectives.

High school physics teacher — Saudi Department of Education | 1990–2008

- Eighteen years of secondary physics teaching, with additional responsibilities in curriculum assessment, laboratory operations, school consultancy and safety planning.
- Longstanding experience translating disciplinary physics knowledge into classroom teaching and learner understanding.

Additional education responsibilities

2000–2008 National curriculum exam marker — National Assignment Centre, Saudi Arabia

2000–2008 School emergency evacuation plans designer — Saudi Department of Education

1994–1997 School consultant — Saudi Department of Education

1990–1999 School laboratory operator — Saudi Department of Education

Research and professional expertise

- Physics and science education; conceptual change research
- Cross-cultural educational research and instrument adaptation
- Quantitative and qualitative data interpretation
- Curriculum analysis; language and culture in science learning
- Conceptual and theoretical framework development
- Early childhood curriculum planning, documentation and reflective practice
- Narrative pedagogy, story-based inquiry and curriculum continuity
- Academic writing, literature synthesis and peer-review response

Additional qualifications

2015 Professional Trainer Licence — Academic Professional Training Association, Saudi Arabia

2005 NLP Diploma — INLPTA, Saudi Arabia

Scholarly identity

Preferred scholarly name: Asmahan Matooq S Al Safwan

Published/indexed name variants: A. Alsafwan; A. Al Safwan; Asmahan Al Safwan

Academic referees and supporting reference letters are available for applications.